



Service Bulletin Number: 4332709

Released Date: 05-feb-2013

Operation of Automotive Natural Gas Engines in Cold Climates

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Satisfactory performance of a natural gas engine in cold and arctic climates requires modification to the engine support systems, operating practices, and maintenance procedures. Everyone involved with equipment in these climates **must** address the fact that cold climates will have an effect on the starting, performance, and reliability of the equipment as well as the engine. The reliability of the engine and equipment will be seriously impaired if **not** properly prepared for the environment in which it is operating. To winterize equipment there are three objectives:

1. Reasonable starting characteristics followed by practical and dependable warm-up of engine and equipment.
2. A unit or installation needs to be as independent as possible from other external influences.
3. Modifications which maintain satisfactory engine operating temperatures with a minimum increase in maintenance of the equipment and accessories.

As the operating climate becomes colder, an increasing amount of equipment modification is required for satisfactory performance. The engines **must** be capable of operating in all climate extremes with satisfactory performance and durability. Some cold climate problems are listed below. While this list contains problems resulting from cold temperatures, it **must** be stressed that additional problems can occur.

⚠ CAUTION ⚠

Do not use ether as engine damage can occur.

Engine Idle

It is **not** recommended that the engine be operated at an idle, no load condition for prolonged periods (above 15 minutes) of time. Operating engines at idle (700 to 900 rpm) in cold ambient temperatures wastes fuel, accelerates wear, and can result in serious engine damage. At low idle (700 to 800 rpm) there will be insufficient coolant flow and coolant temperature at the cab heater to maintain adequate cab heat. In low ambient temperatures, if radiant and convection heat losses are **not** prevented, the engine will **not** consume an adequate amount of fuel to maintain coolant and combustion temperatures. Sufficient support systems are recommended to preclude the necessity to idle engines for prolonged periods.

It is recognized that regardless of recommendations, operators will idle engines for prolonged periods, especially if the driver is sleeping in the cab compartment of the chassis. If the operator insists on prolonged idling, it is recommended that the engine be idled at an rpm which is adequate to maintain coolant temperatures above 60°C [140°F]. In -18°C [0°F], the engine will need to be idled above 1200 rpm. For this type of operation, a full on-off type fan and winter front **must** be used. The thermostat, thermostat seal, and the deaerating system **must not** allow coolant leakage through the radiator core with the thermostat in the closed position.

In 0°C [32°F] temperature, on the ISX12 G CM2180 EJ engine, wrap insulation on the breather inlet tube, outlet tube, and the breather housing. The insulation wrap is available as Cummins® Part Number 2878268.

Electrical Systems

The normal source of energy used to crank an engine is a battery. As the battery temperature decreases, the battery's capacity to produce power is lowered and its ability to recover power is slower. The engine's cranking load increases as the ambient temperature decreases, compounding the battery power loss problem.

Engine manufacturers normally recommend battery and electrical system capacity for specific engine families to provide an adequate start at -18°C [0°F]. For satisfactory operation in colder ambient conditions, the batteries need to be heated to restore their original cranking capacity, or more batteries need to be added. Inadequate battery capacity results in low cranking speed. Cranking speed continues to slow down during the cranking cycle and can progress to the point where the engine will **not** start.

The wire size, length, and connections of the starter circuit determine the ability of the battery to freely transmit power to the starter. Undersized wire acts as a restriction to flow of electrical power. Loose or corroded wires and connections can consume up to 50 percent more of the available battery power before it gets to the starter.

An electric starter simply receives electrical energy and converts it to mechanical energy. The peak efficiency of the starter occurs at engine cranking speeds of between 100 to 150 rpm. At lower speeds the efficiency drops off and current flow increases. If the starter is allowed to stall, and the engine will **not** start, 100 percent of the electrical energy turns to heat and will damage the starter and wiring. If cranking speeds are continuously low, either more energy (battery capacity), higher starter torque, or less cranking load is required.

Fuels

Compressed Natural Gas (CNG)

Natural gas in the tanks will increase in pressure as the ambient temperature rises or decrease in pressure as the temperature falls. The magnitude of the change in pressure is a function of the size of temperature change, starting pressure, and the gas composition.

Liquefied Natural Gas (LNG)

Liquefied Natural Gas (LNG) is produced by purifying natural gas and cooling it to -162°C [-260°F] to turn it into a liquid. Because it must be kept at cold temperatures, LNG is stored in double-walled, vacuum-insulated pressure vessels. LNG is an acceptable fuel, provided the on-board fuel storage and supply system delivers proper pressure, temperature, and complete vaporization to the engine fuel system inlet. The most common way of vaporizing the LNG supply to the engine is through the use of an evaporator, which uses coolant from the engine to convert the LNG to CNG and warm it to acceptable temperatures. If the cooling system does not reach operating temperatures, then the performance of the evaporator may be compromised

Lubricating Oils

Two physical properties, viscosity and pour point, are vital to the lubrication of the engine during cold start up. Viscosity is the **most** important. The lubricating oil **must** pour freely from its containers and it **must** circulate freely throughout the engine. Although it is possible to start an engine when the oil has solidified in the crankcase, doing so can result in catastrophic engine shutdown due to lack of lubrication. There are cold climate operation lubricating oils with a sufficiently low pour point available that will remain in a liquid state at the minimum expected temperatures.

The viscosity of the oil controls engine friction and oil circulation. Oil viscosity influences the following performance factors.

1. Starting and warm up
2. Power output
3. Fuel consumption
4. Engine cooling
5. Starting wear
6. Oil consumption
7. Engine cooling
8. Engine noise.

The first five performance factors are improved by using a low viscosity oil, while the last three factors are improved with a high viscosity oil.

Lubricating oil viscosity is one of the most critical factors in engine startability. If the viscosity is too high, the resistance to cranking ability is too high for the engine to reach an acceptable cranking speed.

It is necessary to heat the engine lubricant before starting the engine if the lubricant that is used does **not** have the proper pour point or viscosity. Heating the lubricant will decrease its viscosity, therefore, increasing lubrication and reducing friction during cranking. All filters **must** be located within the confines of the engine compartment so the heat of the engine can be used to prevent gelling of the lubricants.

Note : White, tan, or brown foam or froth is sometimes found on the underside of the oil fill cap and inside the valve cover. This foam is normal and will not affect the function of the engine.

Air Intake System

There are two aspects of intake air which **must** be taken into consideration when winterizing a vehicle. These considerations are air density and temperature.

Air Temperature

There is a direct correlation between intake air temperature and combustion temperature. One of the end results of a natural gas engine with cold intake temperatures is that combustion temperatures will be reduced.

Below 0°C [32°F], it is recommended that intake air come from inside the engine enclosure. This will provide warm air to the intake and reduce engine heat loss. If the air cleaner is **not** located within the engine compartment enclosure, then ducting **must** be routed from the engine compartment to the air cleaner inlet. Adequate sealing of the engine compartment is necessary. Side curtains alone are **not always** satisfactory. If there are openings around the radiator cowl and underneath the engine, these areas **must** also be sealed. Depending on ambient temperature and engine compartment size, heat the air within the engine compartment to maintain combustion temperatures. An aftercooled engine will perform well in these conditions, as the cooler element, which normally cools heated air down to near the level of the coolant temperature, will warm cold intake air up to near the coolant temperature. It is extremely beneficial to provide a temperature controlled on-off type fan so the radiant and convection heat within the engine compartment is **not** dissipated by air flow.

Cooling System

Coolant within a natural gas engine is primarily used as a heat transfer medium and is subject to freezing in cold and arctic climates. On LNG fuel system, the coolant is used to vaporize and heat the liquefied natural gas to the necessary temperature and pressure to operate the engine.

The lowest freezing temperature that can be provided for the coolant is obtained by using a solution of approximately 60 percent ethylene glycol and 40 percent water. This mixture will turn to slush at about -50°C [-60°F].

There are three basic types of coolant heaters available in the marketplace. The heaters selected **must** maintain the coolant temperature to a minimum of 50°C [120°F] in any temperature encountered.

- Immersion Heaters
 - Immersion heaters have a heating element immersed within the confines of the engine coolant passage. Heat is transferred to the cylinder liners and block via the coolant.

Due to space limitations, the wattage is limited to 2500 watts each or less. An external electrical source is required to operate the heaters.

- Tank Type Heaters
 - Tank type heaters function on the convection principal. That is, a volume of coolant is heated within a small tank and then the heated coolant is circulated through the engine via the hose connections (similar to a percolator coffee pot). There are heaters available from 1000 to 4000 watts. An external electrical source is required for this type of heat. There are also tank type heaters with recirculating pumps which continuously circulate the heated coolant through the engine systems.
- Fuel Burning Coolant Heaters
 - Fuel burning coolant heaters are designed to burn liquid/vapor type fuels and some will operate on a mixture of fuels. The most common fuel burning heaters normally use diesel fuel, propane or gasoline. These are furnace gun type heaters where fuel is ignited via an igniter and the fuel is then sprayed on a transfer medium such as a porous ceramic disc. This flame then heats the coolant via coils contained within the heater and transferred to the engine either by convection or by recirculating pumps incorporated into the heaters. These types of heaters are available in sizes from 6000 to 82000 BTU. Exhaust gases from these heaters can be used to heat engine compartments and accessories such as batteries.

Coolant Flow

The minimum amount of coolant that has to be heated for maximum efficiency of the heaters is the volume of coolant in the block and accessories in the coolant bypass system. To achieve this condition, the thermostats and thermostat seals **must not** allow coolant to flow through the radiator when the thermostat is closed. To check for coolant leakage past the thermostat seal and/or thermostat:

1. Remove the engine outlet hose at the thermostat housing. Hold the hose in the air above the radiator baffle level and run the engine against the governor. If water comes out of the water outlet connection, the thermostat and/or seal is leaking; ninety-seven percent of the time, it is the seal or debris keeping the thermostat open.
2. If the coolant does **not** reach thermostat opening temperature, changing the thermostat will **not** correct the problem.

The second critical area for cold operating engines is coolant flow through the radiator core when the thermostat is in the closed position. A properly functioning deaeration system **must** be incorporated and there **must** be no reverse flow of coolant through either the fill line or the radiator. The deaeration baffle **must not** leak and **must not** have vent holes. To check for this condition:

1. Drain the cooling system.
2. Remove the make up (fill line) at the water pump and plug the end.
3. Fill the area above the baffle with coolant. If water leaks through the drain cock of the radiator, remove the tank and repair. Coolant can be leaking through the core vent tube which needs to be extended in the expansion area.

Radiator Shutters

When they are closed, radiator shutters prevent air flow across the radiator core as well as the engine. Air flow across the engine can extend warm-up time and prevent the coolant from reaching minimum operating temperatures in extremely cold ambient temperatures. The two most commonly used terms associated with the preparation of equipment for low temperature operation are Winterization and Arctic Specifications. Winterization of the engine and/or components so starting and operation are possible in the lowest temperature to be encountered requires:

1. Use of correct materials (Some materials get brittle and fail at low temperatures.)
2. Proper lubrication, low temperature lubricating oils.
3. Protection from the low temperature air (The metal temperature does **not** change as a result of air flow, but the rate of heat dissipation is significantly increased by the flow of cold air over the exposed surfaces.)
4. Proper external heating source available
5. Electrical equipment capable of operating in lowest expected temperature.

Arctic Specifications reference the design material and specifications of components necessary for satisfactory engine operation in extremely low temperatures to -54°C [-65°F]. Contact Cummins Inc., or the equipment manufacturer, to obtain the special items required. The Arctic Specifications listed on the following pages are only general guidelines.

Fan Drive

A temperature controlled fan drive can be used to reduce cold air circulation at the engine to help conserve compartment heat. These devices function by completely or partially disconnecting the fan from its drive source at a specified temperature. Several types of fan drives are commonly available: viscous, thermatic, and thermo-modulating.

Equipment Preparation

The amount of modification varies as the temperature changes. Therefore, the following tabulation of recommended equipment is grouped into three temperature ranges. When equipment heaters are required as recommended in this bulletin, it is suggested that all system heaters (coolant, oil, and intake air) be used. The following recommendations apply to all engine models, except as noted.

Winterize 0 to -23°C [32 to -10°F]

- Use ethylene glycol antifreeze to protect to -29°C [-20°F]
- Radiator shutters
- System to provide a recommended 150 rpm cranking speed at -23°C [-10°F]
- Provide for warmed intake air from engine compartment in temperatures below -9°C [15°F]
- Shorten breather tube to at least 12 inches above oil pan flange on highway applications for operations below -18°C [0°F]

- For the ISX12 G CM2180 EJ engine, wrap insulation on the breather inlet tube, outlet tube, and the breather housing. The insulation wrap is available as Cummins® Part Number 2878268
- Coolant heaters
- Oil heaters

Winterize -23 to -32°C to [-10 to -25°F]

- Use ethylene glycol antifreeze 50 percent, water 50 percent
- Radiator shutters sealed around edge and adjusted to close tight or shutters plus radiator cover
- Winterized heavy duty 24 volt cranking system to provide a recommended 150 rpm cranking speed at -32°C [-25°F] after preheat
- Heated intake air using intercooler manifold, engine compartment enclosures with air cleaner intake air drawn from engine compartment
- Dry type air cleaner
- Shorten breather tube to at least 30 cm [12 in] above oil pan flange
- For the ISX12 G CM2180 EJ engine, wrap insulation on the breather inlet tube, outlet tube, and the breather housing. The insulation wrap is available as Cummins® Part Number 2878268
- Coolant heaters
- Oil heaters
- OFF/ON type fan
- Oil pan shield for on-highway units
- Battery warmer
- Heavy duty arctic-type batteries
- Battery cables size per Table 8
- All filters to be mounted in the engine compartment. Before exposing the equipment to the low temperature, change all filters, drain the systems, and flush thoroughly with arctic lubricants
- Insulate all coolant and oil lines exposed to the elements.

Arctic Specifications -32 to -54°C [-25 to -65°F]

- Use ethylene glycol antifreeze 60 percent, water 40 percent mixture
- Tight winter front cover for radiator in front of shutters
- Winterized heavy duty 24 volt cranking system to provide a recommended 150 rpm cranking speed at -32°C [-25°F] after preheat
- Provide for heated intake air using intercooler manifold, engine compartment enclosures with air cleaner intake air drawn from the engine compartment
- Dry type air cleaner. Install in engine compartment
- Shorten breather tube to at least 30 cm [12 in] above oil pan flange
- For the ISX12 G CM2180 EJ engine, wrap insulation on the breather inlet tube, outlet tube, and the breather housing. The insulation wrap is available as Cummins® Part Number 2878268
- Coolant heaters
- Oil heaters
- OFF/ON type fan

- Insulated cover for lower part of the engine compartment including oil pan shields
- Battery warmer
- Heavy duty arctic type batteries
- Battery cables size per Table 8. Hold circuit wiring short as possible
- All filters to be mounted in the engine compartment. Before exposing the equipment to the low temperature, change all filters, drain the systems, and flush thoroughly with arctic lubricants. Replace the filter a second time after the flushing operation
- Repack fan hub and water pump with arctic grease.

Winterize mechanical controls and instruments:

- Tachometer
- Tachometer cable
- Starting aid control cable
- Pressure gauges
- Temperature gauges.
- All regular lubricants for cables to be removed and left dry or replaced with arctic type lubricants
- Provide auxiliary engine compartment space heat sufficient to bring engine temperature to -32°C [-25°F]
- Use -54°C [-65°F] electrical equipment and wiring
- Use -54°C [-65°F] hose
- Use -54°C [-65°F] rubber in the filter and dipstick expanders
- Use -54°C [-65°F] belts
- Insulate all coolant and oil lines exposed to the elements. Insulate or cover with heat strip tape all coolant and oil lines exposed to the elements.

⚠ CAUTION ⚠

Anti-leak antifreezes are not recommended for use in Cummins® engines. Although these antifreezes are chemically compatible with DCA water treatment, the anti-leak agents can clog the coolant filters, oil coolers and radiators, making them ineffective.

Maintenance

The accessory equipment used for low temperature operation and starting require maintenance on a regular schedule like other components on the engine. Poor or unscheduled maintenance **always** results in loss of productive time. Normal maintenance practices **must** be followed on all components. Maintenance suggestions for all engines are:

Winterize 0 to -23°C [32 to -10°F]

- Adhere to a lubricating oil change schedule necessary to maintain clean oil.
- Check electrical systems daily.
- Check cold weather support systems weekly.

Winterize -23°C to -32°C [-10°F to -25°F]

- Adhere to an oil change schedule necessary to maintain clean oil. Check level every two hours.
- Check electrical systems daily.
- Check cold weather support systems weekly.
- Check temperature controls, thermostat, and shutters weekly.

Arctic Specifications -32 to -54°C [-25 to -65°F]

- Adhere to an oil change schedule necessary to maintain clean oil. Check level every two hours.
- Check electrical systems daily.
- Check cold weather support systems weekly.
- Check temperature controls, thermostat, winter front, space heater starters, and starter electrical system weekly.

Note : All Temperature Categories: In areas of blowing snow, it will be necessary to remove the air cleaner element and remove packed snow if the air cleaner inlet is located outside the engine compartment

Accessory Equipment

Coolant Heaters

Electric or fuel fired coolant heaters are an excellent starting aid. Warm coolant creates a more satisfactory starting condition by reducing cylinder to piston friction, where most cold weather cranking drag occurs. Also, more complete combustion occurs when the combustion chamber is warmed.

Electric Coolant Heaters

There are two types of electric coolant heaters, direct immersion and external tank. Figure 1 shows a typical direct immersion type coolant heater. Figure 2 shows a typical external tank type heater.

Tank Type Heaters

⚠ CAUTION ⚠

Do not operate the external type heaters with the engine running.

⚠ CAUTION ⚠

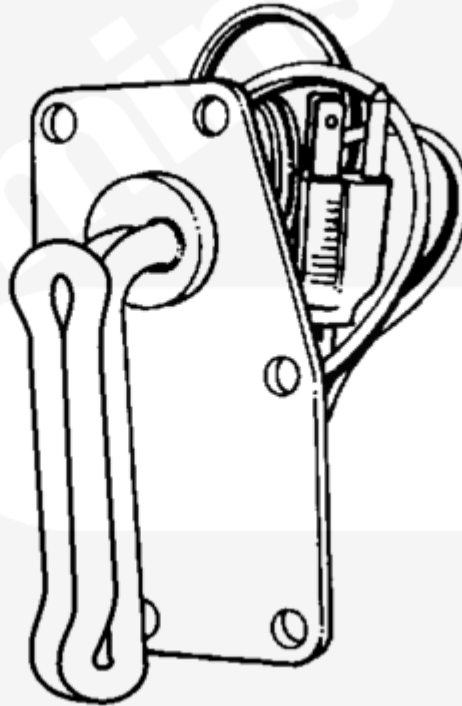
Circuit breakers and fuses are slow to act in extreme low temperatures; therefore, electrical accessories must have adequate size wiring and equipment.

Note : Some tank type coolant heater assemblies include a check valve and thermostat. An oil pressure switch for automatic cutoff of the heating element must be used with all heater assemblies. An appropriate pressure switch and relay assembly is included with the 460 volt or 3 phase supply system.

Some tank type coolant heaters are of the external type, mounted horizontally. The Kim Hot Start models have an exclusive flapper type reinforced neoprene check valve. The valve forces the heated coolant to move in only one (outlet) direction, and also stops a bypass to normal coolant circulation while the engine is running.

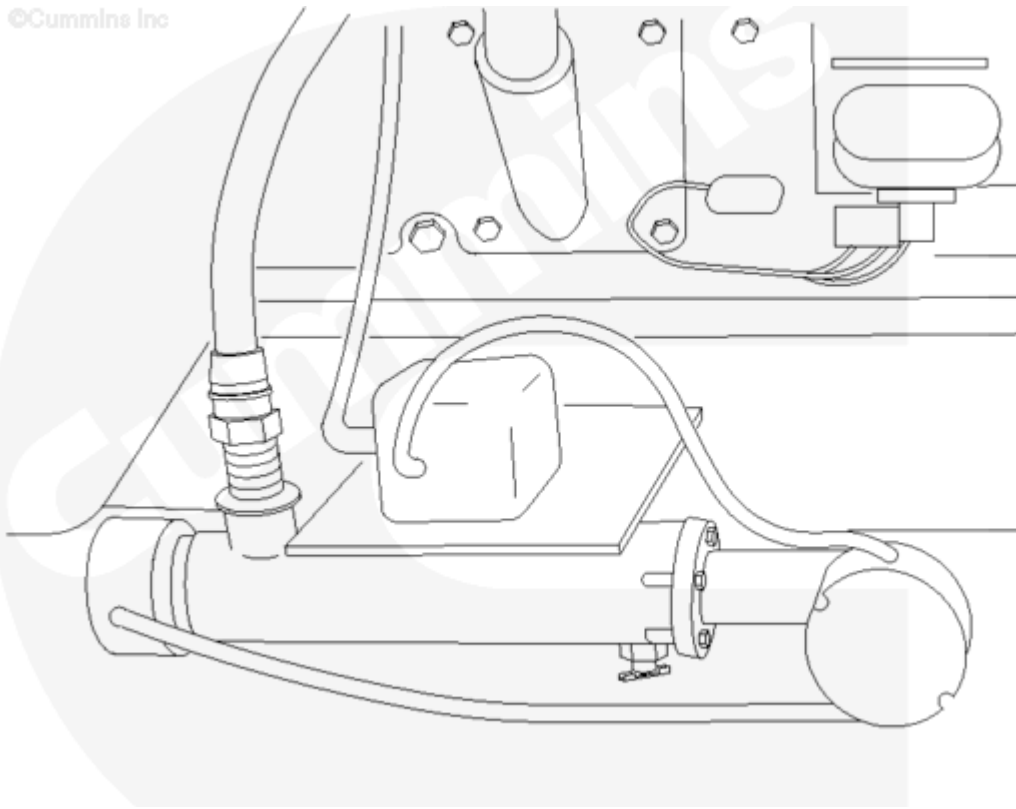
Figure 2 illustrates a tank type external coolant heater that can be applied to all engines. Connect the heater inlet to the lowest point of coolant (water pump or lower radiator tank) and the heater outlet to a point on the engine block higher than the heater inlet to obtain the best thermo-syphon water flow.

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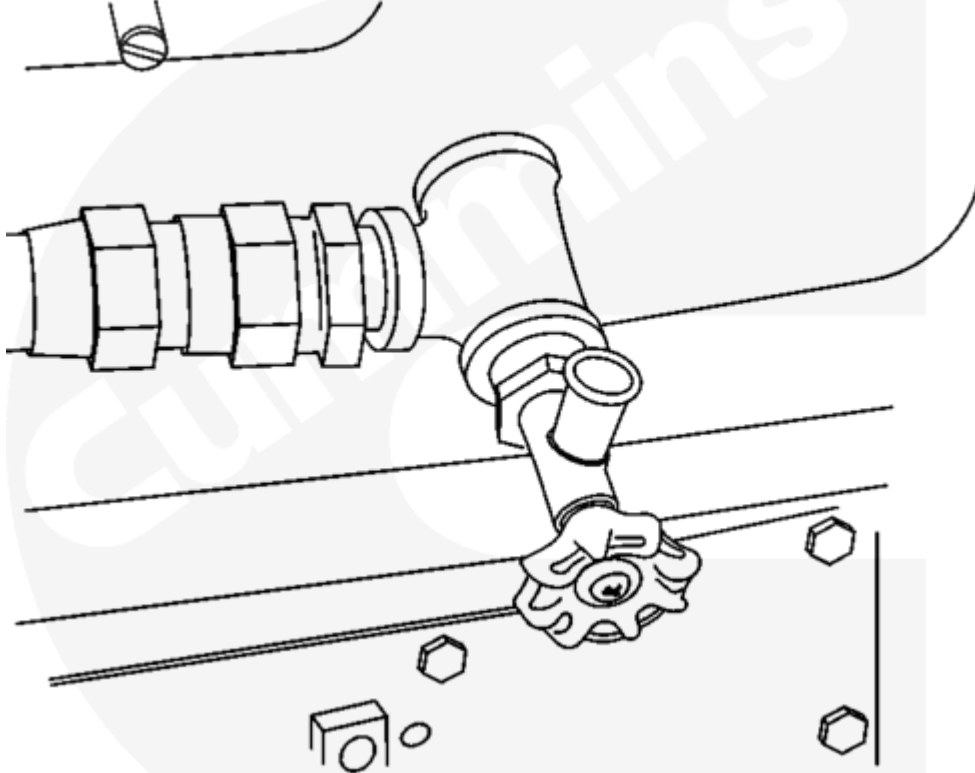
coldfg1

Figure 1: Direct immersion electric coolant heater.



coldfg2

Figure 2: External tank type electric coolant heater.



coldfg3

Figure 3: Coolant heater outlet to cylinder block.

These heaters are versatile because they can be used for space heating of the engine compartment, the battery, or the cab, in addition to the coolant housing.

Table 1: Immersion Type Coolant Heaters				
Engine Model	Option	Part Number	Description	Wattage/Voltage
B Gas, BLPG	CH 9010	3905113	Cable, Engine Heater	1000/120
		3918597	Heater, Engine	
	CH 9042	3923050	Cable, Engine Heater	750/120
		3975628	Heater, Engine	
	CH 9045	3913112	Cable, Engine Heater	750/120
		3975629	Heater, Engine	
	CH 9046	3913106	Cable, Engine Heater	750/120
		3975628	Heater, Engine	
	CH 9048	3905114	Cable, Engine Heater	750/240
		3975629	Heater, Engine	
	CH 9049	3913112	Cable, Engine Heater	750/240
		3975629	Heater, Engine	
	CH 9051	3905113	Cable, Engine Heater	1000/120
		3918597	Heater, Engine	
	CH 9053	3905114	Cable, Engine Heater	1000/120
		3918598	Heater, Engine	
B Gas Plus, B LPG Plus	CH 9010	3905113	Cable, Engine Heater	1000/120
		3918597	Heater, Engine	
	CH 9045	3913112	Cable, Engine Heater	750/120
		3975629	Heater, Engine	
C Gas, C Gas Plus	CH 9010	3905113	Cable, Engine Heater	1000/120
		3918597	Heater, Engine	
C Gas Plus - International	CH 9068	3905113	Cable, Engine Heater	1000/120
		4935057	Heater, Engine	

Table 1: Immersion Type Coolant Heaters				
Engine Model	Option	Part Number	Description	Wattage/Voltage
L Gas Plus	CH 9068	3905113	Cable, Engine Heater	1000/120
		4935057	Heater, Engine	
	CH 9068	3905113	Cable, Engine Heater	1000/120
		4935057	Heater, Engine	
L10GI	CH 2001	3027513	Element, Heating	1500/120
	CH 2013	3026156	Washer, Plain	1500/120
		3819413	Element, Heating	
		3905113	Cable, Engine Heater	
L10GII, L10GIII	CH 2001	3027513	Element, Heating	1500/120
ISLG	CH 9068	3905113	Cable, Engine Heater	1000/120
		4935057	Heater, Engine	
	CH 9095	4935057	Heater, Engine	1000/120
ISX12 G	CH 2043	3681360	Cable, Engine Heater	1500/120
		3903990	Screw,Hex Flange Head C	
		4965497	Heater, Engine TLA	
Replacement elements are available through Cummins Filtration™.				

Table 2: Tank Type Coolant Heaters without Electrical Junction Box, Thermostat, and Check Valve		
Heater Cummins Filtration™ Part Number	Voltage	Wattage
259269	115	1000
259270	115	1000
259271	230	1000
259272	230	1000
259288	230	4000

Table 2: Tank Type Coolant Heaters without Electrical Junction Box, Thermostat, and Check Valve		
Heater Cummins Filtration™ Part Number	Voltage	Wattage
259289	460	4000

Table 3: Thermostat for Heaters Listed in Table 2	
Cummins Filtration™ Part Number	Temperature
259290	27° to 38° C [80° to 100° F]
259290	38° to 49° C [100° to 120° F]
259293	49° to 60° C [120° to 140° F]
259294*	49° to 60° C [120° to 140° F]
259295	60° to 71° C [140° to 160° F]
259296*	27° to 38° C [80° to 100° F]
259368*	60° to 71° C [140° to 160° F]
* Explosion Proof	

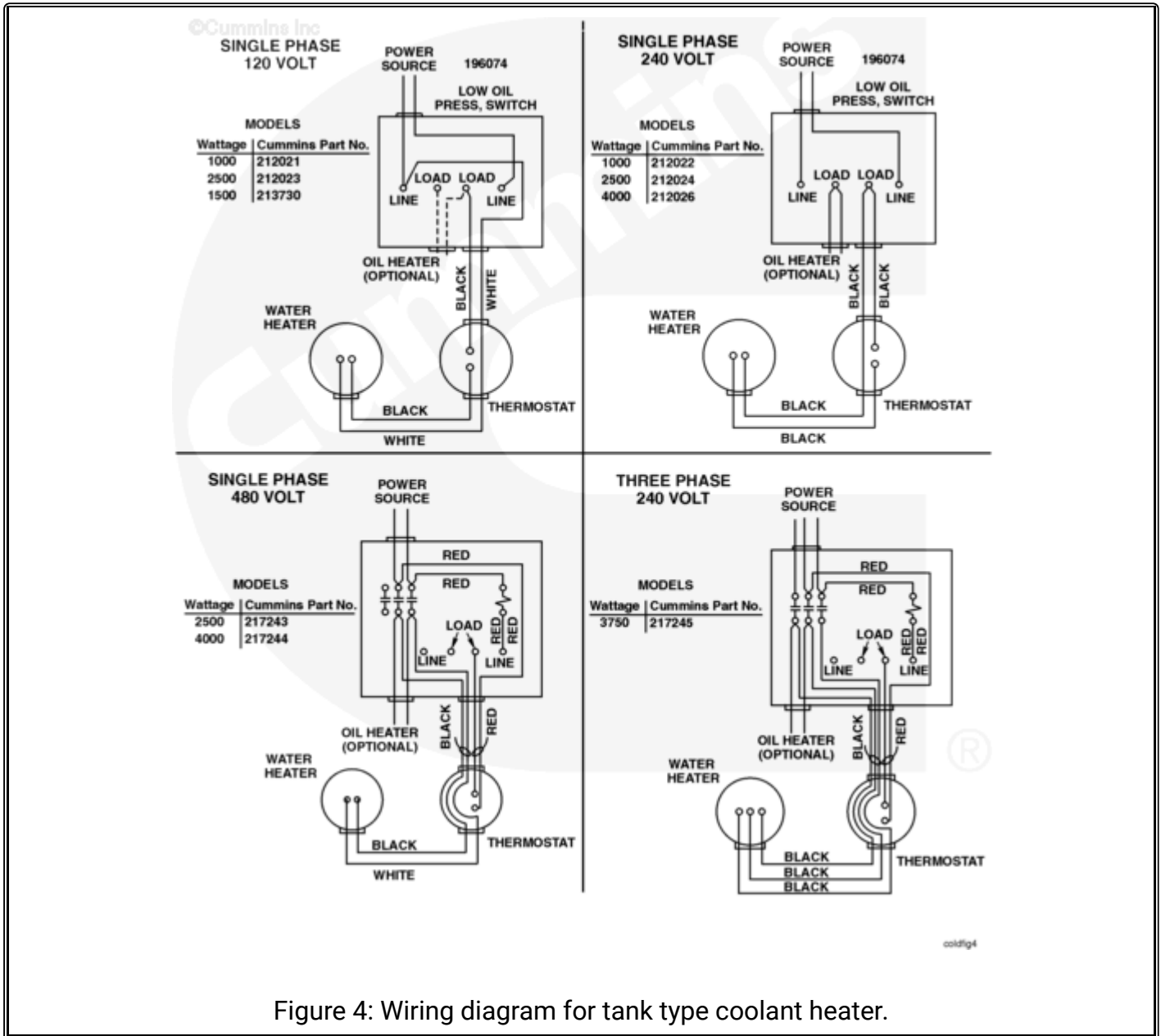
Table 4: Tank Type Coolant Heater with Electrical Junction Box, Thermostat, and Check Valve				
Cummins® Part Number	Voltage	Wattage	Wiring Diagram Number	Low Oil Pressure Switch
212021	120	1000	1	196074
212022	240	1000	2	196074
213730	120	1500	1	196074
212023	120	2500	1	196074
212024	240	2500	2	196074
217243	480	2500	3	-
212026	240	4000	2	196074
217244	480	4000	3	-
217245	240 (3-phase)	3750	4	-

Use a minimum hose size Number 16. For increased coolant flow and greater heater efficiency, use Number 20 hose.

⚠ CAUTION ⚠

Use Teflon™ type hose.

Table 5: Suggested Wattage for Water and Oil Heaters		
Temperature Range	Medium Duty	Heavy Duty
0 to -18°C [32 to 0°F]	750 or 1000 watt immersion, 1000 watt coolant tank type, 150 watt oil immersion	1000 watt coolant immersion, 1500 watt coolant tank type, 150 watt oil immersion
-18 to -32°C [0 to -25°F]	1000 watt coolant immersion, 1000 watt coolant tank type, 150 watt oil immersion	1500 watt coolant immersion, 2000 watt coolant tank type, 300 watt oil immersion
-32 to -51°C [-25 to -60°F]	1500 watt coolant immersion, 1500 watt coolant tank type, 300 watt oil immersion	2250 watt or 2500 watt coolant tank type, 300 watt oil immersion



Fuses for Coolant Heaters

- 115 Volts

- 20 ampere fuse - maximum wattage 2250
- 25 ampere fuse - maximum wattage 2875
- 30 ampere fuse - maximum wattage 3450
- 230 Volts
 - 20 ampere fuse - maximum wattage 4600
 - 25 ampere fuse - maximum wattage 5750
 - 30 ampere fuse - maximum wattage 6900

Battery Box Heating - All Applications

Battery heating can be accomplished with electrical power or hot fresh air from such sources as the cab, fuel fired cab heater, hot exhaust air, and hot liquid from the engine coolant system.

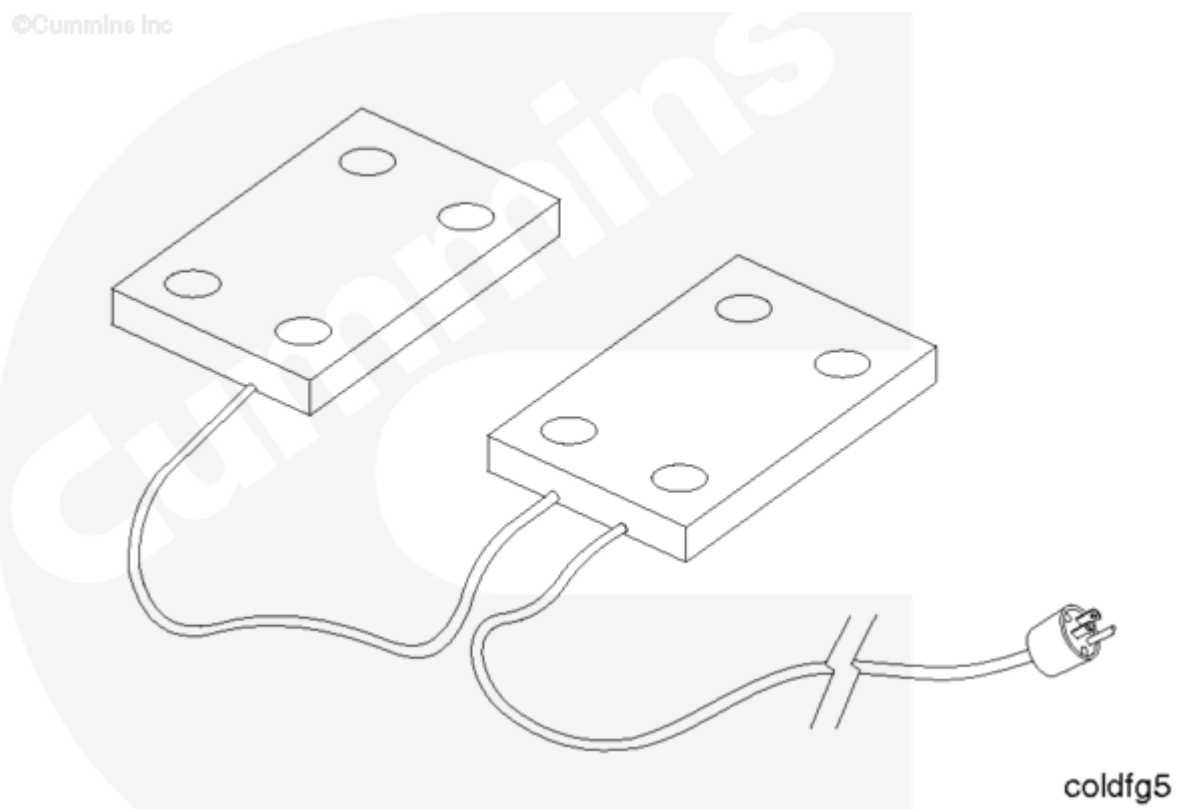


Figure 5: Electric battery warmer.

Electric Battery Heaters

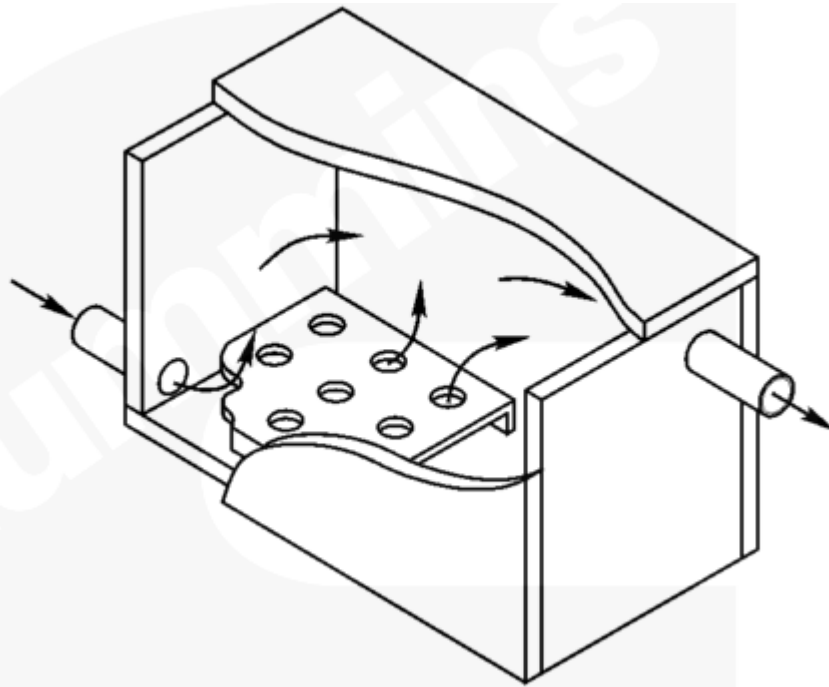
Heat resistor strips can be mounted to the side of the battery box. The elements should be arranged to distribute heat around the batteries, avoiding hot spots. This strip heater is available as Cummins® Part Number 143519, rated at 120 volts, 285 watts. Part Number 143519 can be obtained from Cummins Inc., Columbus, Indiana, U.S.A. 47201.

A typical battery warmer pad is available as Cummins Filtration™ Part Number 259262, rated at 150 volts, 200 watts. See Figure 5. Part Number 259262 can be obtained from Cummins Filtration™ Division, Cummins Inc., Cookville, Tennessee, 38501.

Battery temperature **must not** exceed 58°C [125°F]. Use external electrical power as available; this type load on the batteries would soon drain them.

Note : In all types of heaters, an automatic thermostat control must be provided to prevent overheating of the battery.

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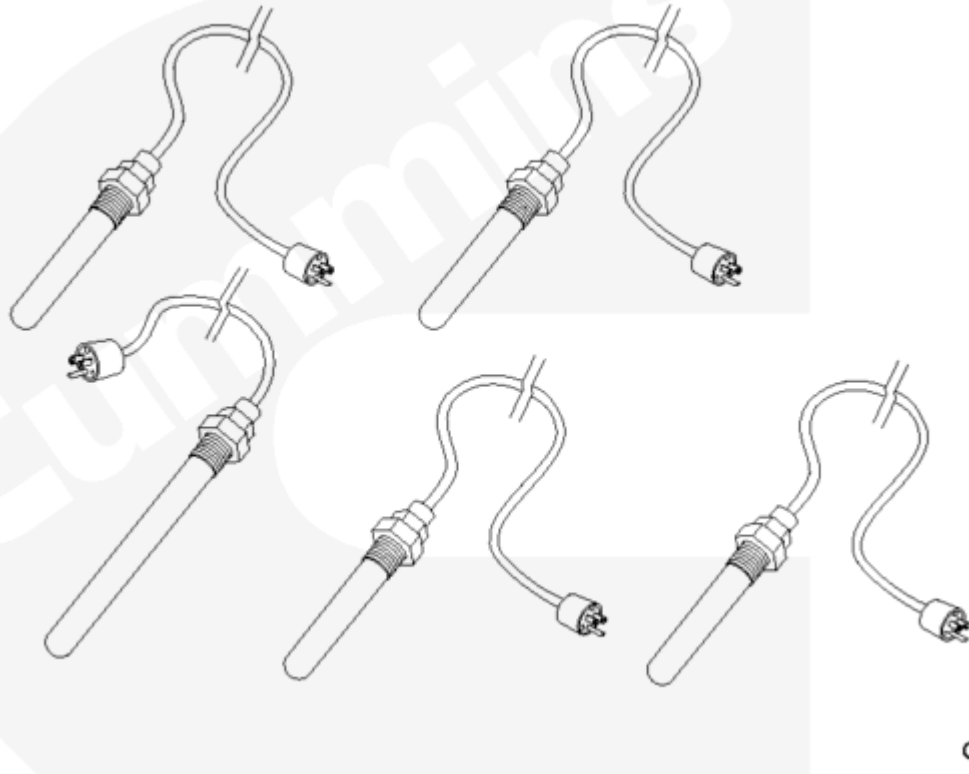


coldfg6

Figure 6: Heated battery box.

Hot Fresh Air Heating

The battery box can be heated by warm air from the cab or from a fuel-fired cab heater. The warm air is piped into the lower part of one side of the battery box and discharged at the upper part of the other side of the box. It is advisable to provide 25.4 mm to 50.8 mm [1 in to 2 in] clearance on each side of the battery for circulation. Further aid to heating can be provided by mounting the battery off the floor of the box through use of a corrugated or perforated support to permit circulation of warm air on all sides of the battery. Insulate the box.



coldfg7

Figure 7: Shielded type lubricating oil heaters.

Lubricating Oil Heaters

Lubricating oil heating is generally accomplished with immersion type electrically powered heaters. Typical shielded type lubricating oil heaters are shown in Figure 7. All models have stainless steel sheath elements inserted into separate aluminum wells.

Note : Installation instructions are supplied for mounting and wiring each unit.

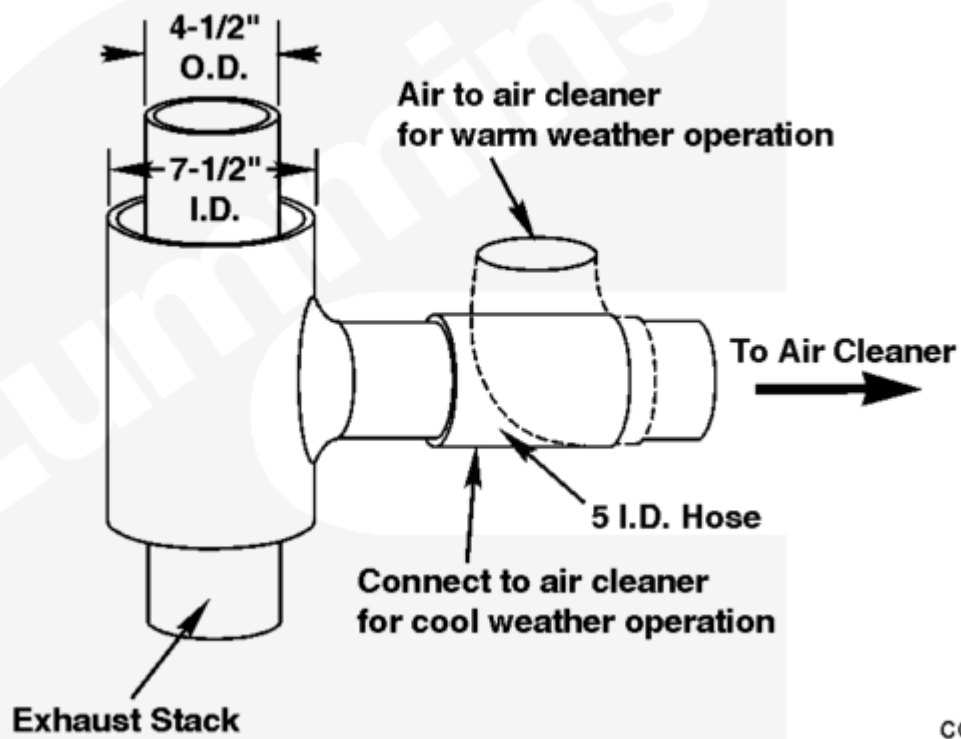
Table 6: Oil Immersion Heater Applications

Oil Immersion Heaters Part Number	Voltage	Wattage	Length of Heating Element	Hole Diameter	Oil System Capacity
251920	115	150	194 mm [7-5/8 in]	25 mm [1 in]	7.6 to 15 liters [2 to 4 Gal]
256855	115	150	194 mm [7-5/8 in]	12.7 mm [1/2 in]	7.6 to 15 liters [2 to 4 Gal]
259254*	115	150	133 mm [5-1/4 in]	12.7 mm [1/2 in]	7.6 to 15 liters [2 to 4 Gal]
259254*	230	150	133 mm [5-1/4 in]	12.7 mm [1/2 in]	7.6 to 15 liters [2 to 4 Gal]

Table 6: Oil Immersion Heater Applications					
Oil Immersion Heaters Part Number	Voltage	Wattage	Length of Heating Element	Hole Diameter	Oil System Capacity
251921	230	200	194 mm [7-5/8 in]	25 mm [1 in]	15 to 30 liters [4 to 8 Gal]
259256*	115	300	133 mm [5-1/4 in]	12.7 mm [1/2 in]	15 to 30 liters [4 to 8 Gal]
259257*	230	300	133 mm [5-1/4 in]	12.7 mm [1/2 in]	15 to 30 liters [4 to 8 Gal]
259258*	115	450	216 mm [8-1/2 in]	12.7 mm [1/2 in]	30 to 45 liters [8-12 Gal]
259259*	230	450	216 mm [8-1/2 in]	12.7 mm [1/2 in]	30 to 45 liters [8-12 Gal]
* These heaters are supplied with an adapter for brazing into a one (1) inch hole.					

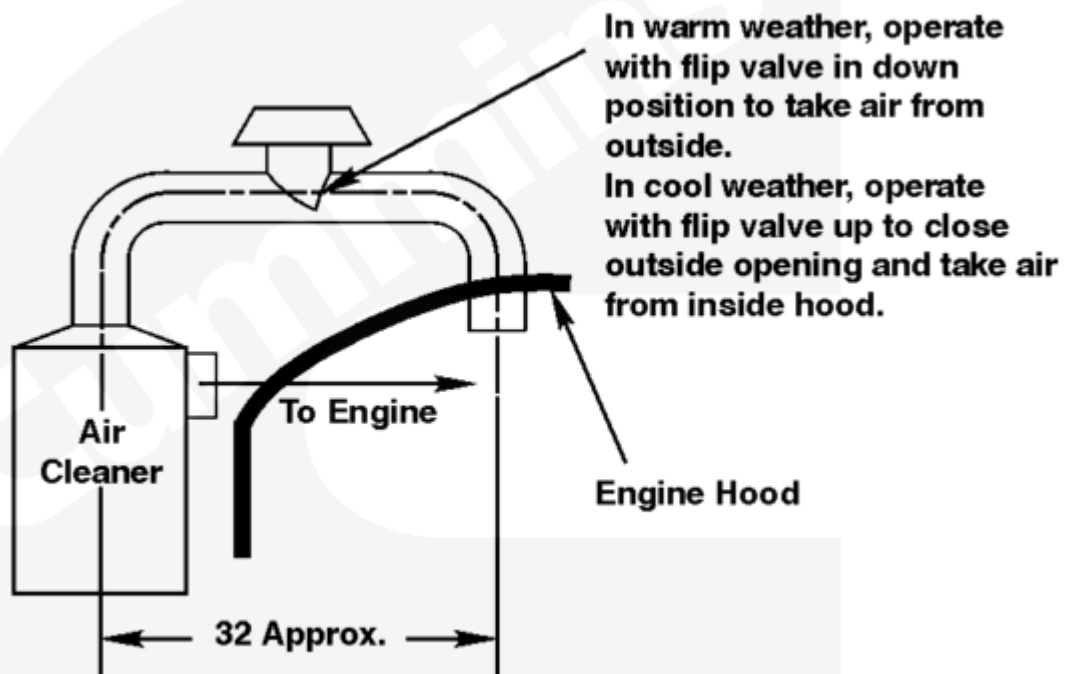
Table 6A: Optional Thermostat Part Number	
Part Number	Fits the Following Oil Immersion Heaters
259261**	259254, 259256, 259258
3009521	259255, 259257, 259259
** Thermostat recommended in moderate temperature areas if wattage exceeds 10 watts/quart of oil in the lubricating system.	

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coldfg8

Figure 8: Utilizing exhaust heat to warm intake air.



coldfg9

Figure 9: Exterior mounted air cleaner with flip valve.

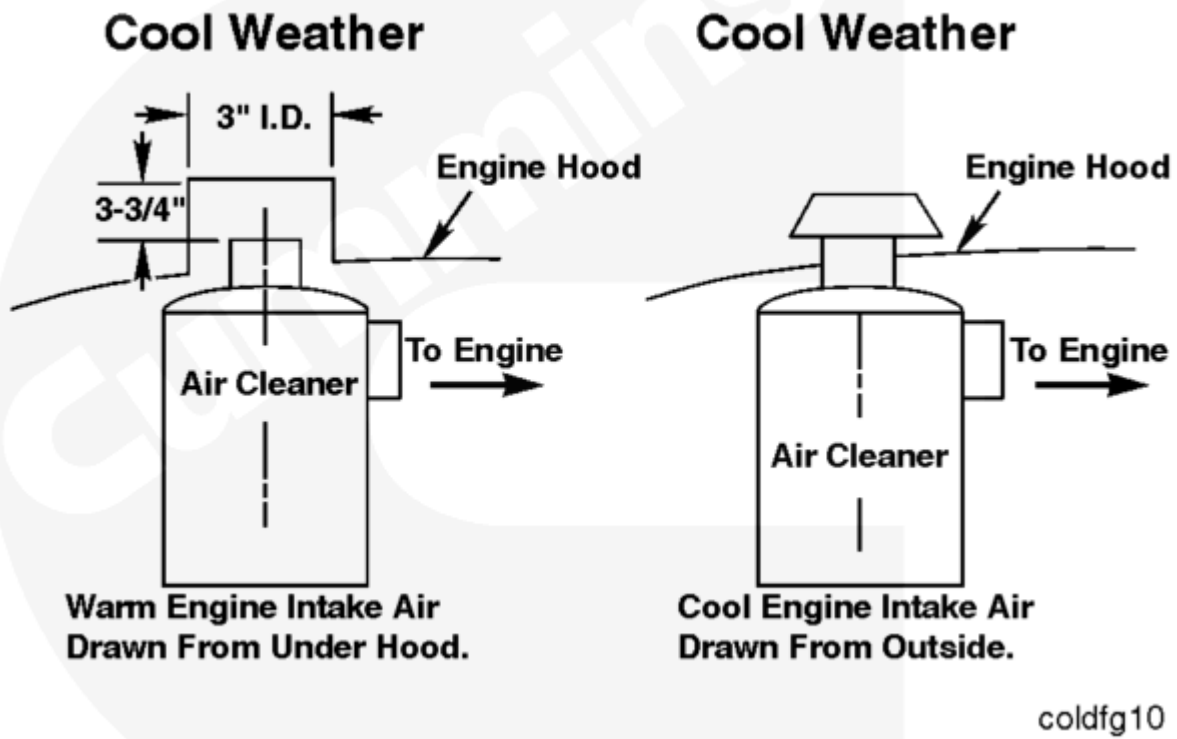


Figure 10: Utilizing engine heat to warm intake air.

Engine Intake Air Heating

Engine intake air heating can be accomplished by pulling air from around the exhaust pipe (Figure 8), out of the engine compartment (Figures 9 and 10), or through an aftercooler. It is **always** desirable and necessary under constant operation in ambient temperatures below -32°C [-25°F] to preheat the air, especially at lighter loads to provide conditions leading to complete combustion. Normally, air pulled from the enclosed engine compartment will be preheated sufficiently for operation to approximately -32°C [-25°F]. At lower temperatures, build a compartment. (Figure 8) around the exhaust stack and pull air through the jackets to the air cleaner. Regardless of the type of arrangement used to pull air from the engine compartment and/or exhaust jacket, provision **must** be made to pull outside air into the engine intake for operation at higher temperatures. This recommendation applies to all engine models.

Note : In heavy snow and ambient temperatures below -9°C [15°F], use the intake design shown in Figure 9 and Figure 10 to pull air from under the hood to prevent plugging of the air filter

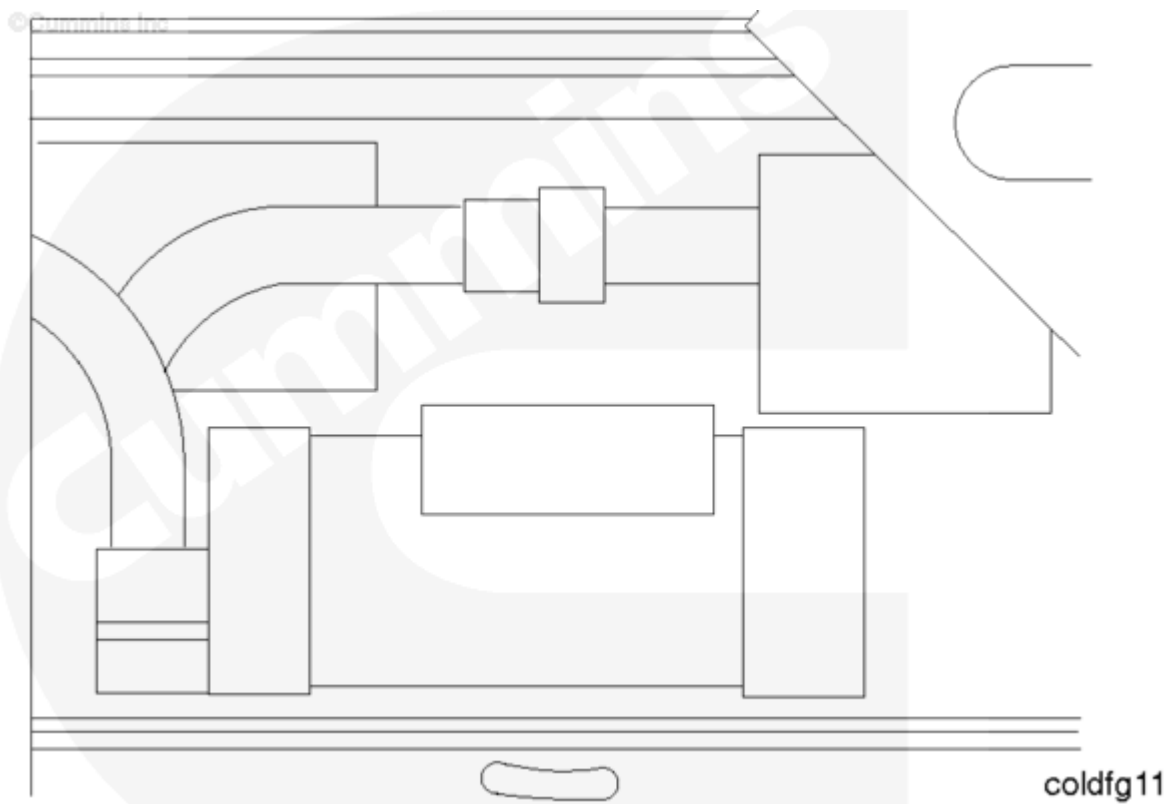


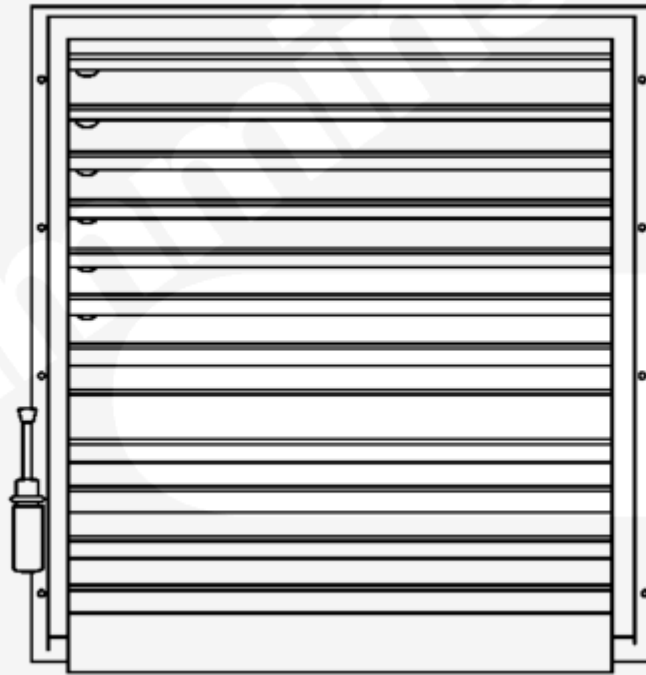
Figure 11: Typical space heater for engine compartment enclosure.

Engine Compartment Heaters

When the temperature drops below -32°C [-25°F], it is desirable to add heat to the engine compartment in addition to the coolant, batteries, and lubricating oil, to bring the engine temperature up to a minimum of -32°C [-25°F] so satisfactory starting is realized. A typical space heater, applicable to all engine models, is shown in Figure 11.

⚠ WARNING ⚠

Ether or ether starting aids must never be used on natural gas engines.



coldfg16

Figure 12: Radiator cover with variable opening.

Insulation Materials

The engine compartment requires consideration because the walls and panels **must** be sealed and insulated so no interference is created with engine components. It **must** be fire safe and permit sufficient air entrance so the engine intake air restriction is under 38 cm [15 in] water with a clean air filter. The air feed to the compartment **must** be in an area of clean air and as far away from engine air intake ducting as possible. The insulation material **must** be oil impervious to provide fire-proof protection. Suitable materials are available in strips or sheets for this application. Use approximately 25.40 to 38.10 mm [1 to 1 1/2 in] thickness insulation mounted securely by straps so it can **not** fall on the engine or into the fan. Gypsum, glass foam, or mineral wool will furnish sufficient insulation protection.

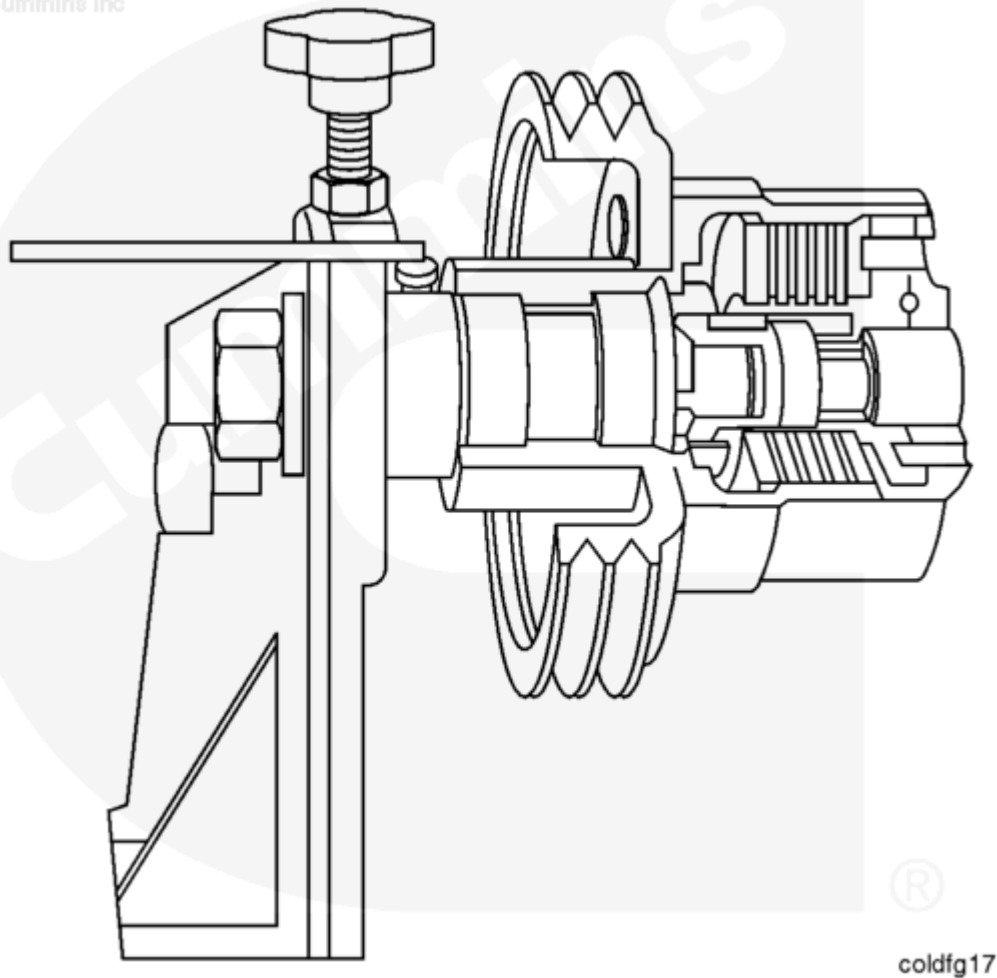


Figure 13: Thematic fan drive.

Fan Drive

A temperature controlled fan drive will reduce air circulation at the engine and help conserve compartment heat. These devices function by disconnecting the fan from its drive source at a specified temperature. Several types of drives are available. The thermo-modulating drive is air operated and provides gradual engagement depending on coolant temperature. The viscous drive utilizes a fluid coupling and will vary fan speed between 40 percent and 90 percent of its drive source speed.

Note : At low idle and in cold ambient temperatures, the viscous fan drive can stay engaged due to the viscosity of the fluid.

Radiator Covers

A winter front cover, which is more air tight than shutters, is used to seal off the engine compartment and radiator at -32°C [-25°F] or lower temperatures. Winter front covers are available from most automotive supply stores. Kysor and Cadillac, radiator shutter manufacturers, can supply special front covers. Generally these are custom made to fit the application and **must** be air tight. The cover opening is adjusted manually, depending on the ambient temperature and operation of the shutters providing control of air to the radiator.

Oil Pan Covers

Waterproof covered insulation or a heavy canvas blanket can be suspended under the engine or the cover can be a metal plate. These parts are available from any canvas or sheet metal shop and **must** be custom fit to the vehicle.

Electrical Equipment

Cranking Motor

The cranking motor **must** be of heavy duty construction to withstand the longer cranking periods at higher amperage loads and **not** overheat; use 24 volts or 32 volts. All starters **must** have the positive pinion engagement feature and **must** be winterized for lowest expected temperatures.

Cranking Motor Cables

Use starting system cable size and length as specified in Table 7. If a frame ground is used, scrape paint and grease from the point of cable connection, and tighten securely to provide good electrical connection.

Table 7: Cable Sizes (Total Length Allowable in Cranking Motor Circuit - AWG Gauge)						
Circuit Voltage	Maximum Voltage Drop	Maximum Resistance	Number 00	Number 000	Number 0000 or Two Number 0	Two Number 00
12-VDC	0.12V/100 amps	0.0012 Ohm	Up to 3.6 m [up to 12 ft]	3.6 to 4.8 m [12 to 16 ft]	4.8 to 6 m [16 to 20 ft]	6 to 7.6 m [20 to 25 ft]
12-VDC High Output Starting Motor	0.075V/100 amps	0.00075 Ohm	Up to 2 m [up to 7 ft]	2 to 2.7 m [7 to 9 ft]	2 to 2.7 m [7 to 9 ft]	3.6 to 4.2 m [12 to 14 ft]
24-VDC to 32-VDC	24-VDC to 32-VDC	0.002 Ohm	Up to 6 m [up to 20 ft]	6 to 8.2 m [20 to 27 ft]	6 to 8.2 m [20 to 27 ft]	10.6 to 13.7 m [35 to 45 ft]

Note : Two strands of Number 0 cable can be used in place of one Number 0000 cable provided that all connections are carefully made, so current in each parallel cable will be equal.

Note : The cross sectional circular mil area of Number 0000 cable is double that of Number 0 cable.

As a final check, measure the voltage drop and ampere flow in the system while cranking the engine for circuit resistance.

Table 8: Wire Sizes	
Circuit Description	AWG Gauge Size
Battery - Starter cables	See Table Number 0
Battery charging generator/alternator output leads (total length of output leads to be under 4.8 m [16 ft]); See sheet 3 of diagram 194076	-
Sensing lead battery voltage-regulator (alternator only)	10
All control circuit wiring, including generator/alternator field lead, in diagram 194076.	14
Accessory lead	As required by load

Batteries

Heavy duty construction with ample reserve power **must** be specified for all batteries used under low temperature starting conditions.

SAE J573h Battery Ratings

Cold Cranking Ampere Rating (CCA); the requirements for this standard are that the battery be cold soaked until the center of the cell reaches -18°C [0°F] and then discharged for 30 seconds at the rate which will provide at least 1.2 volts for each cell at the end of this time. This discharge rate indicates the cold cranking amperes. See Table 9 and 10. Use 1,280 specific gravity electrolyte (Federal specifications O-S-801) in all 6TN and 8TN batteries used in low temperatures.

Minimum Battery Capacity for Cummins® Engines

Waterproof covered insulation or a heavy canvas blanket can be suspended under the engine or the cover can be a metal plate. These parts are available from any canvas or sheet metal shop and **must** be custom fit to the vehicle.

Table 9: Minimum Battery Capacity for Cummins Engines in Winter Climate; -18°C [0°F]
Minimum Engine Temperature

		12-VDC		24-VDC		30-VDC	32-VDC
Engine Series	Engine Displacement (Liters)	Cold Cranking Amperes -18°C [0°F]	Reserve Capacity	Cold Cranking Amperes -18°C [0°F]	Reserve Capacity	Cold Cranking Amperes -18°C [0°F]	Cold Cranking Amperes -18°C [0°F]
B	5.88	1280	480	640	240	-	-
C	8.3	1800	640	900	320	645	-

Table 9: Minimum Battery Capacity for Cummins Engines in Winter Climate; -18°C [0°F] Minimum Engine Temperature							
		12-VDC		24-VDC		30-VDC	32-VDC
Engine Series	Engine Displacement (Liters)	Cold Cranking Amperes -18°C [0°F]	Reserve Capacity	Cold Cranking Amperes -18°C [0°F]	Reserve Capacity	Cold Cranking Amperes -18°C [0°F]	Cold Cranking Amperes -18°C [0°F]
L	8.9	1280	480	640	240	-	-
L10	10	1800	640	900	320	800	780
ISL	8.9	1280	480	640	240	-	-
ISX	11.9	1800	640	900	320	800	780

Table 10: Minimum Battery Capacity for Cummins Engines in Mild Climate; 0°C [32°F] Minimum Engine Temperature							
		12-VDC		24-VDC		30-VDC	32-VDC
Engine Series	Engine Displacement (Liters)	Cold Cranking Amperes 0°C [32°F]	Reserve Capacity	Cold Cranking Amperes 0°C [32°F]	Reserve Capacity	Cold Cranking Amperes 0°C [32°F]	Cold Cranking Amperes 0°C [32°F]
B	5.88	1280	480	640	240	-	-
C	8.3	1280	480	640	240	475	-
L	8.9	1280	480	640	240	-	-
L10	10	1280	480	640	240	475	455
ISL	8.9	1280	480	640	240	-	-
ISX	11.9	1280	480	640	240	945	520

Hydraulic or Air Cranking Motors

Hydraulic or air cranking motors can be used where the oil or air reservoirs and pressures are available.

⚠ CAUTION ⚠

Due to the initially high cranking speeds from these type cranking motors, always prelubricate the engine and/or turbocharger in temperatures below -12°C [10°F].

Personnel Protection

The wind chill chart, Table 11, is included in this bulletin to emphasize the protection required for the operator working under low temperature conditions. They **must** be protected adequately to correctly perform the task of starting and operating the equipment. Wind chill affects humans more than metals, since it is actually a measure of evaporation. The sensation to the skin is a lower temperature as given on the chart. As an example, assume an ambient temperature of -32°C [-25°F] with a 48.27 km [30 mile]/hour wind, the equivalent chill temperature is -61°C [-77°F] or it is equivalent to -61°C [-77°F] with no wind velocity. Temperature in this range will freeze exposed human flesh in less than one minute. Metal would remain at -32°C [-25°F].

Table 11: Wind Chill Rate

Wind Velocity mph	Temperature °F															
	30	25	20	15	10	5	0	-5	-10	-15	-20	-25	-30	-35	-40	-45
0	30	25	20	15	10	5	0	-5	-10	-15	-20	-25	-30	-35	-40	-45
10	20	14	8	2	-4	-10	-15	-21	-27	-33	-39	-45	-51	-56	-62	-68
15	13	7	0	-6	-12	-18	-24	-30	-36	-42	-48	-54	-60	-66	-72	-78
20	9	2	-5	-12	-19	-26	-32	-38	-44	-50	-56	-62	-68	-74	-80	-86
25	5	-2	-9	-17	-24	-31	-37	-43	-49	-55	-61	-67	-73	-79	-85	-91
30	3	-5	-12	-20	-27	-34	-40	-46	-52	-58	-64	-70	-76	-82	-88	-94
35	0	-7	-14	-22	-29	-36	-42	-48	-54	-60	-66	-72	-78	-84	-90	-96
40	-1	-9	-16	-24	-31	-38	-44	-50	-56	-62	-68	-74	-80	-86	-92	-98

Temperature and wind velocity related in terms of wind chill showing equivalent temperature under still conditions. (Values are approximate and do **not** allow for varying humidity.)

Highway Operation

The following equipment or equivalent protection **must** be provided for highway applications operating in arctic climates. See Pages 1, 2 and 3.

- Provide for warm intake air in ambients below -9°C [15°F].
- Use a cover over the radiator such as a winter front.
- Shield oil pan and lower engine compartment from air blasts created by velocity of vehicles.
- Use a thermatic controlled fan.
- Heat the lubricating oil when the engine is **not** operating.
- Heat the coolant when the engine is **not** operating.
- Exposed lines **must** be insulated.
- Filters, pumps, or reservoirs outside the engine compartment **must** be insulated to protect against ambients and air velocity.
- Provide a separate heater directed to operator's feet to maintain driver and cab temperatures.
- Warm the batteries.
- Use an alcohol vaporizer to protect compressed air system.
- If the engine is exposed behind the cab, a cover **must** be provided to enclose the engine in the compartment.

Document History

Date	Details
2013-2-5	Module Created
2013-7-23	The initial edits were driven by FIRG issue 36869, which found that insulation was needed on the ISX12 G crankcase breather inlet tube in arctic conditions. However, other minor changes were requested and added during the cross functional review.
2014-2-28	Update underhood temp to reflect recommenations in 107-101-004 (graphic 00200040)

Last Modified: 22-Apr-2014
